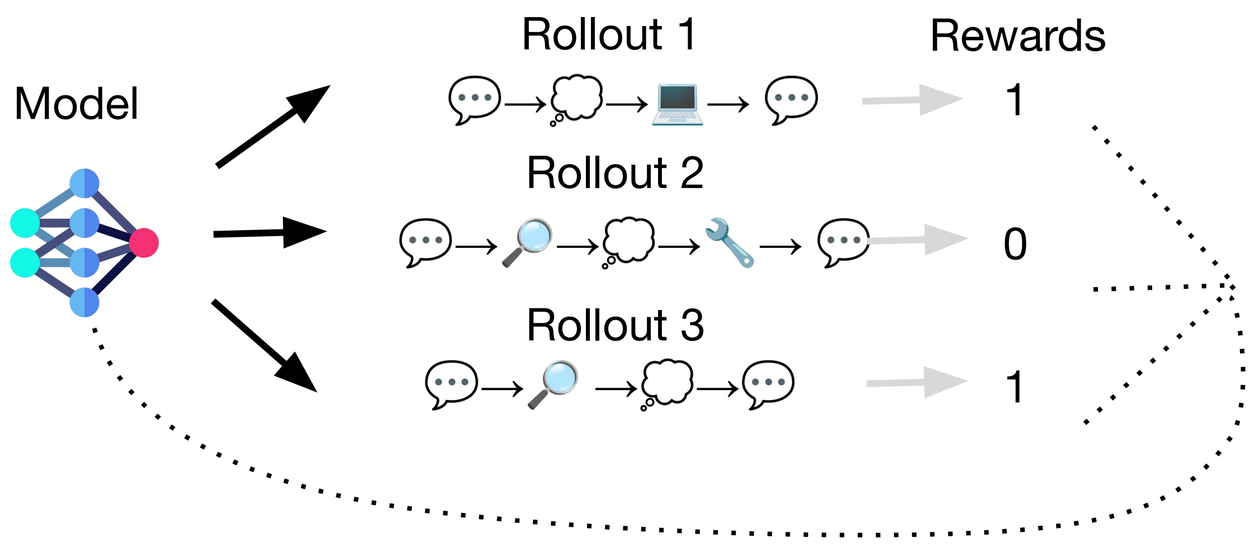
GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning

Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, Omar Khattab

Motivation



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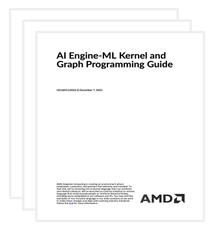

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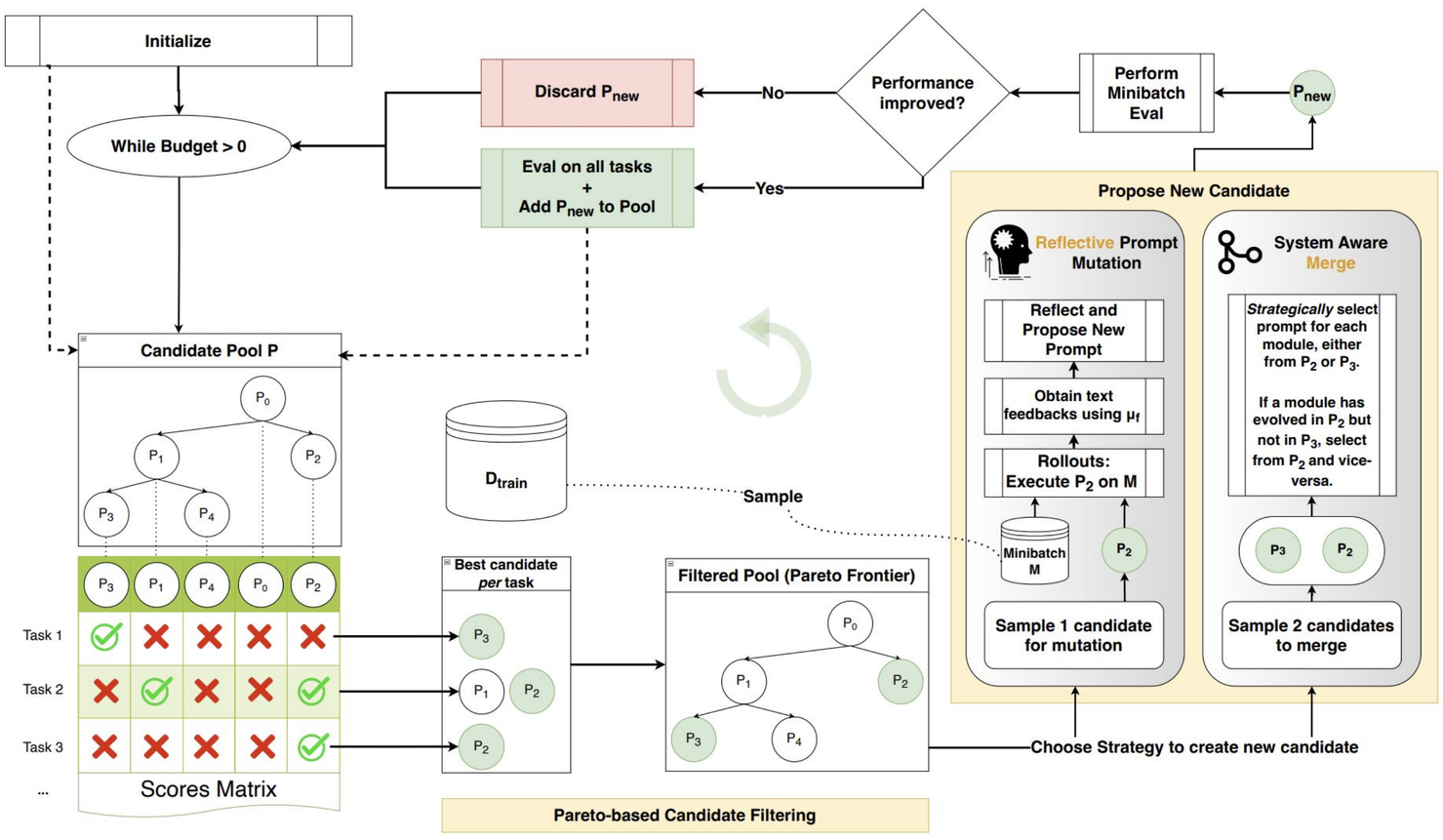
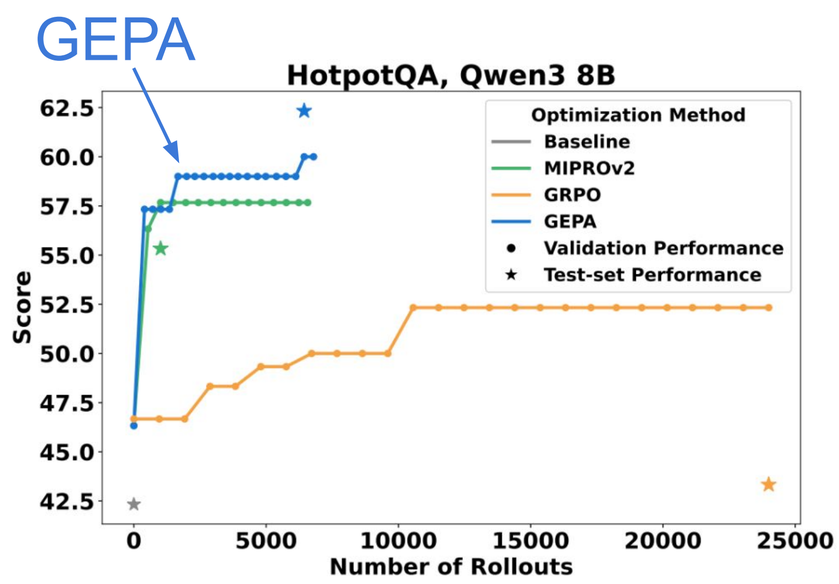
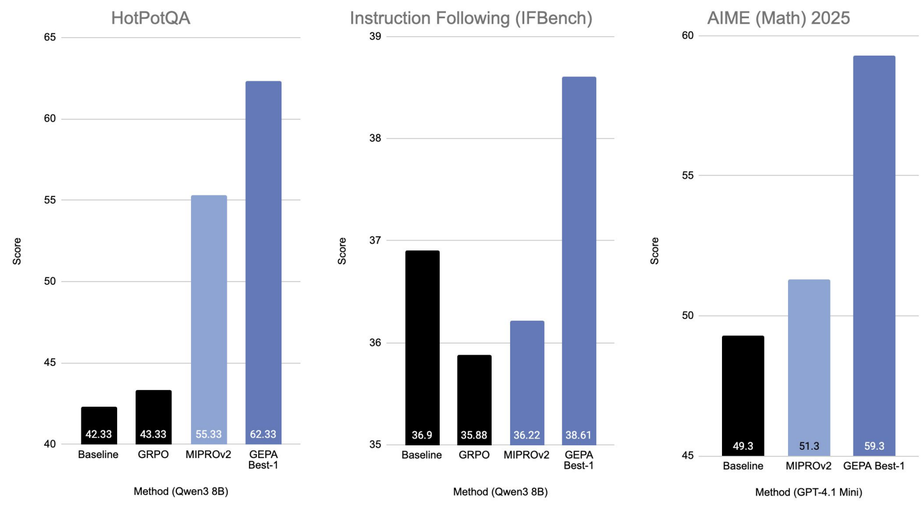
Expensive rollouts: either the LLM workflow for the task or the task metric is slow/expensive to execute

GEPA: Reflective Evolution and Pareto-based Candidate Selection



Results: GEPA is highly sample efficient!





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Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23	—
GRPO	43.33	35.88	38.67	86.66	**38.00**	51.26	48.91	+3.68
MIPROv2	55.33	36.22	47.33	81.55	20.00	46.60	47.84	+2.61
GEPA	62.33	**38.61**	**52.33**	**91.85**	32.00	**51.95**	**54.85**	**+9.62**
GEPA+Merge	**64.33**	28.23	51.67	86.26	32.00	**51.95**	52.40	+7.17
Total optimization budget (# rollouts)								
GEPA (+Merge)	6871	3593	7051	2426	1839	1839	3936	—
GRPO	24000	24000	24000	24000	24000	24000	24000	—

GPT-4.1 Mini	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	38.00	47.79	46.33	78.57	49.33	58.20	53.03	—
Trace (OptoPrime)	60.33	51.19	46.00	74.18	45.33	60.74	56.30	+3.27
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GEPA-Qwen-Opt	65.67	49.83	54.67	90.05	52.67	59.31	62.03	+9.00

GEPA generated prompts are incredibly detailed!

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You will be given two input fields: `question` and `summary_1`. Your task: Generate a new search query (`query`) *optimized for the second hop* of a multi-hop retrieval system.

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Purpose and Context:

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Key Observations and Lessons:

- First-hop documents often cover one entity or aspect.
- Remaining relevant documents often involve connected or higher-level concepts mentioned in `summary_1` but not explicitly asked in the original question. The `query` should target these *missing*, but logically linked, documents.
- Avoid merely paraphrasing the original question or restating known facts from `summary_1`.
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- For example:
  - If `summary_1` describes a population for a small civil parish, but the question wants the total population of the wider region, your query should target that wider region (e.g., "Madeira archipelago population in 2011").
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How to Build the Query:

- Identify entities or topics mentioned in `summary_1` that are related but different from first-hop documents.
- Reframe the query to explicitly mention these broader or related entities *connected to the original question.*
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Practical Strategy:

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Output:

- Produce `query` as a clear, concise question or keyword phrase designed for efficient retrieval of second-hop documents.
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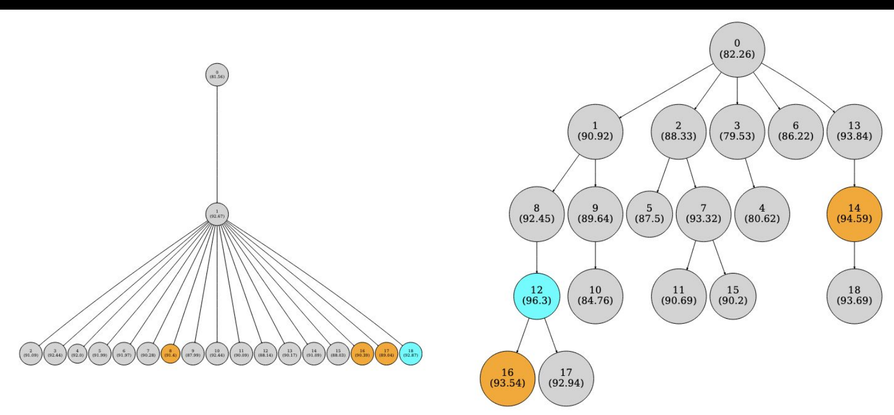
Pareto-based Candidate Selection is Important



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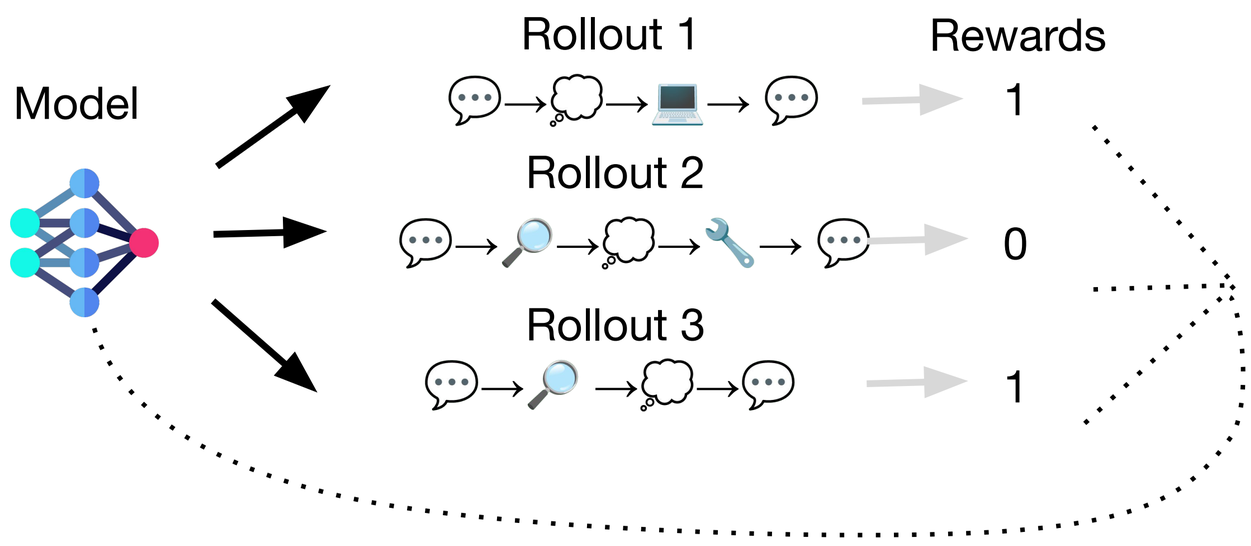

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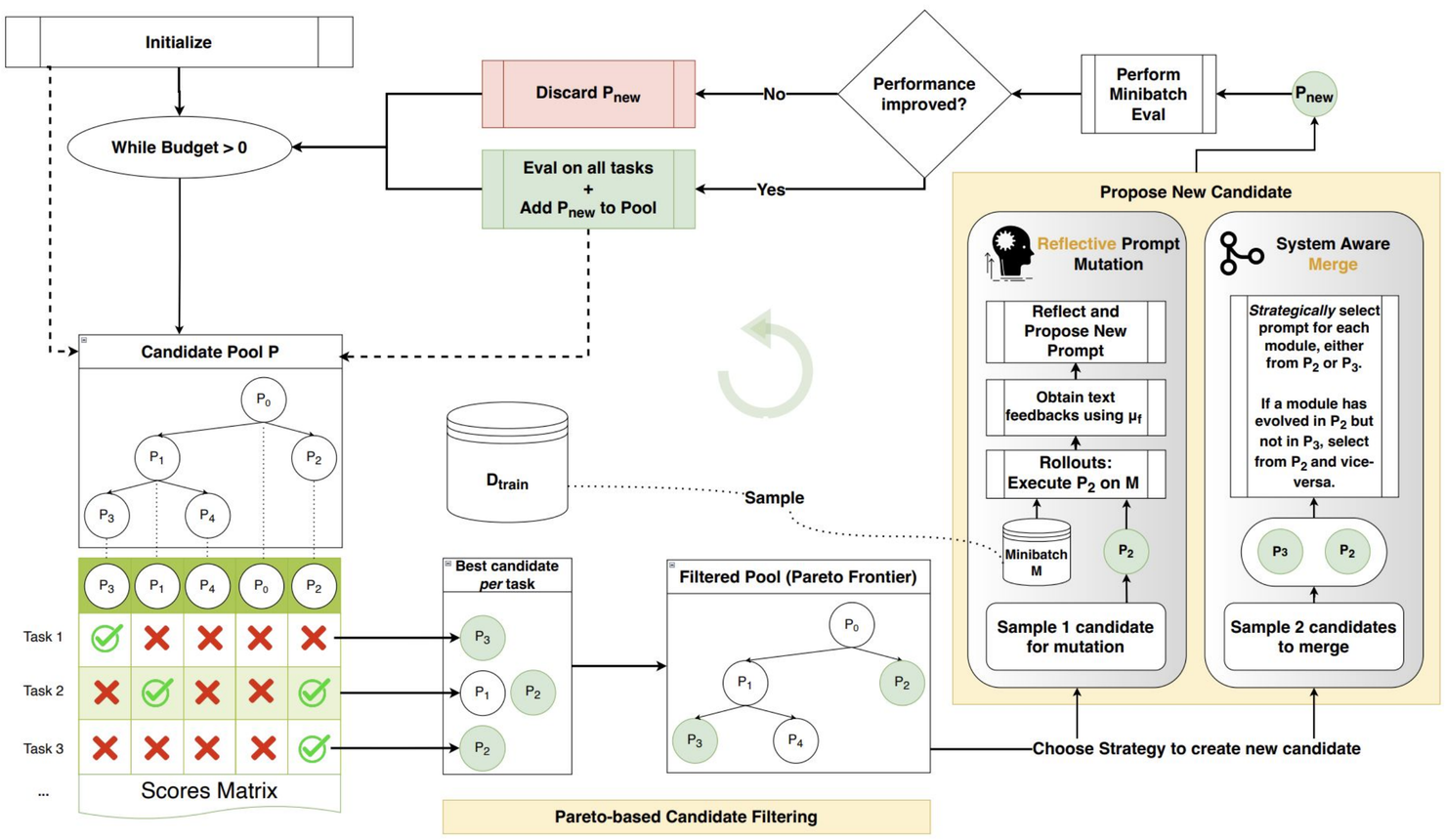

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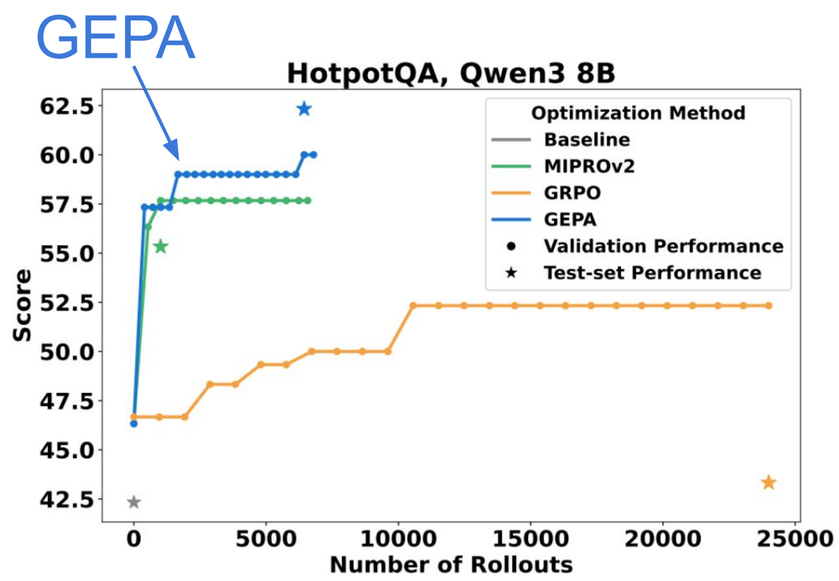

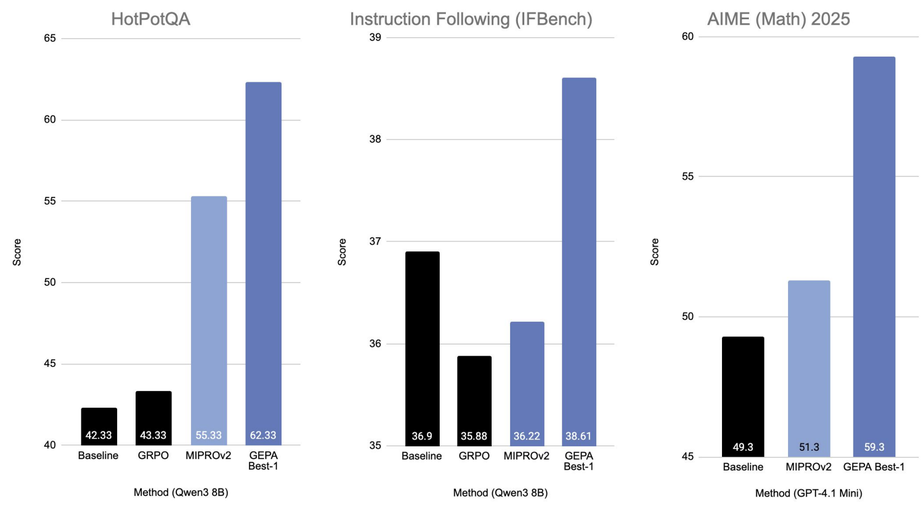

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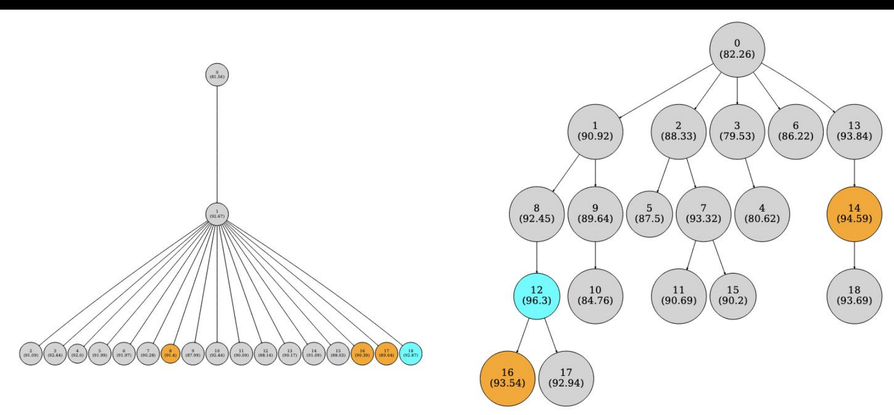

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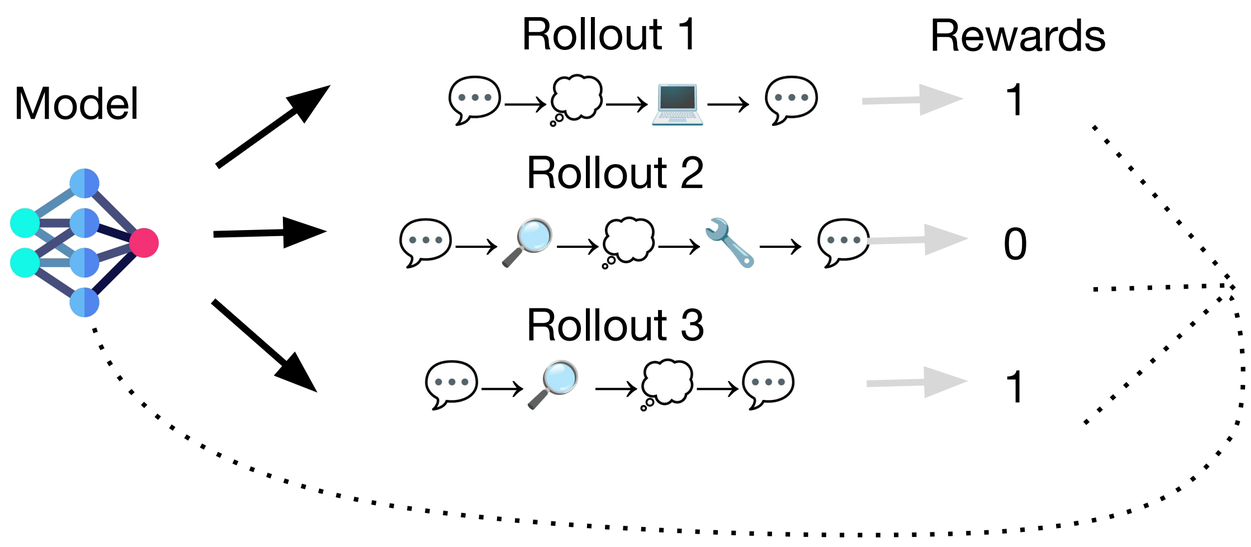

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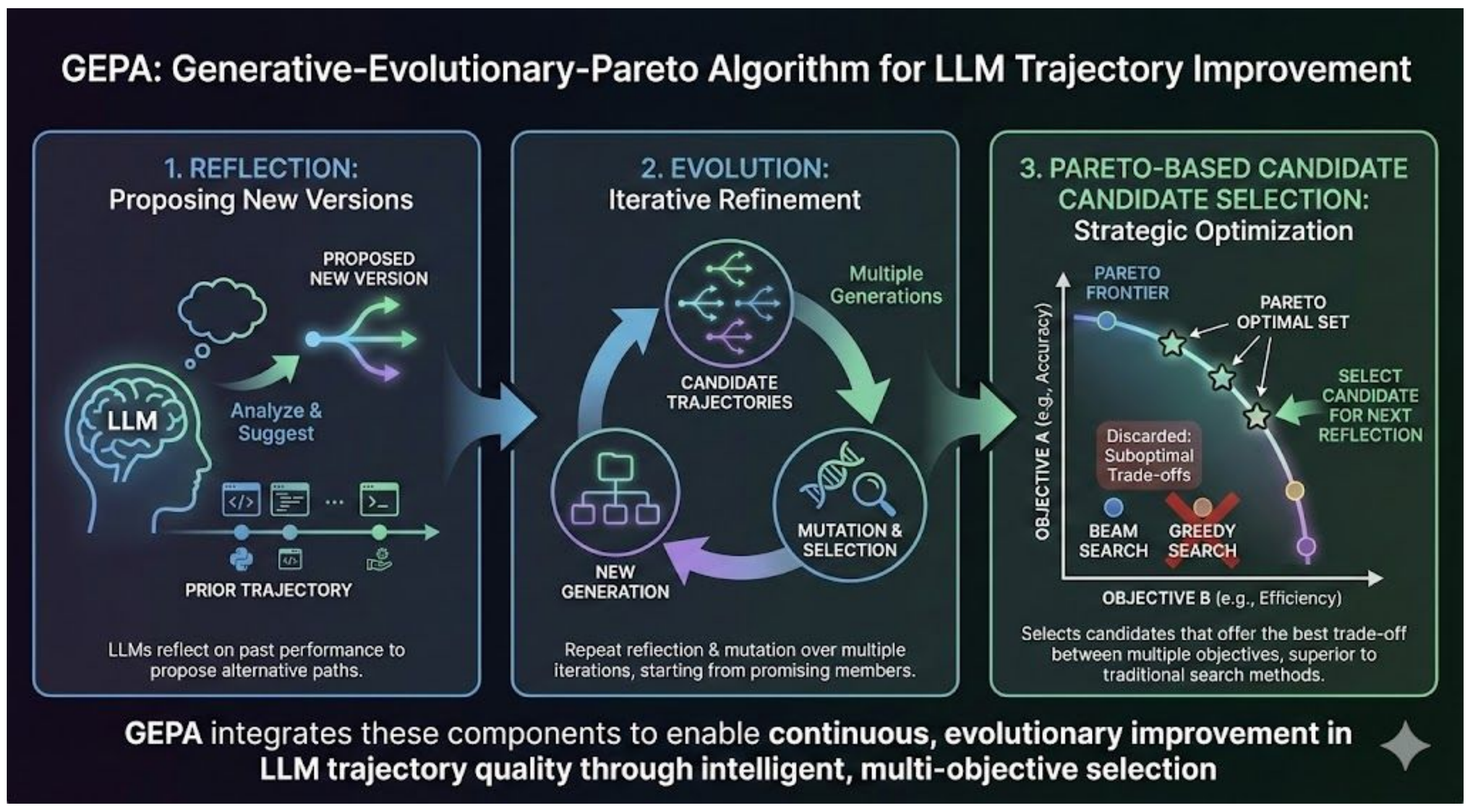

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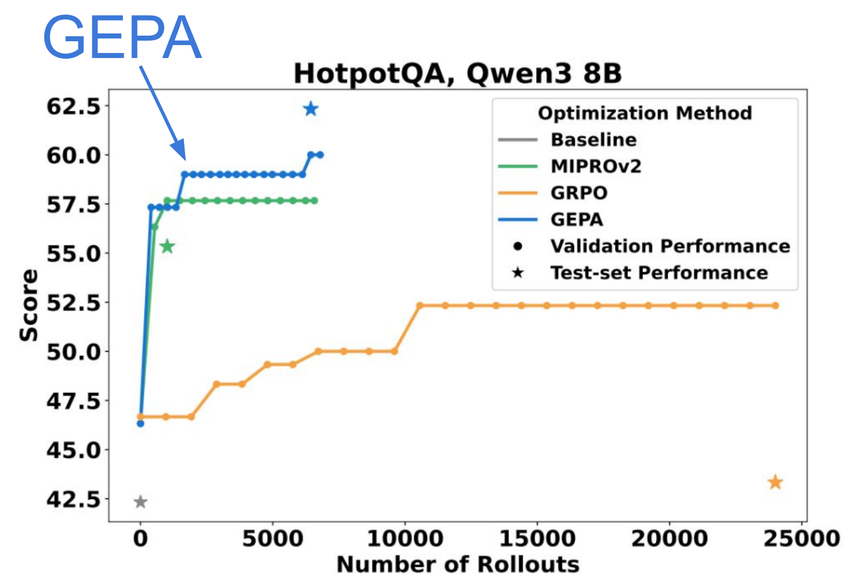

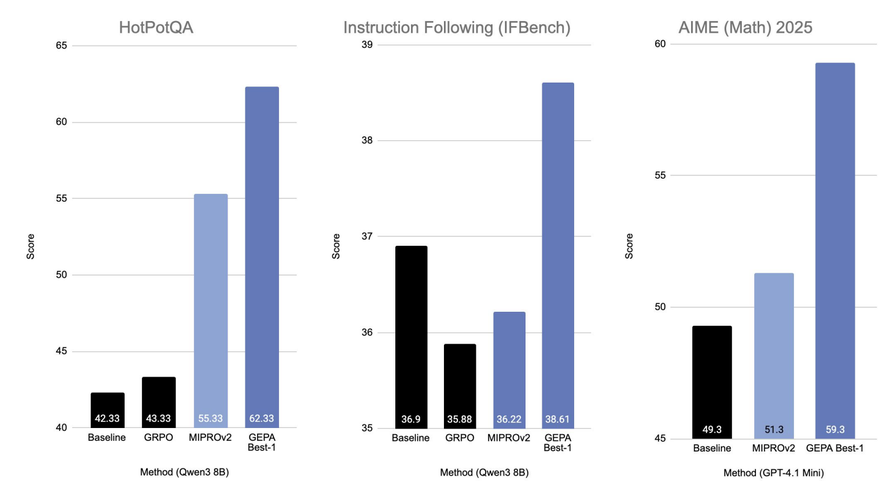

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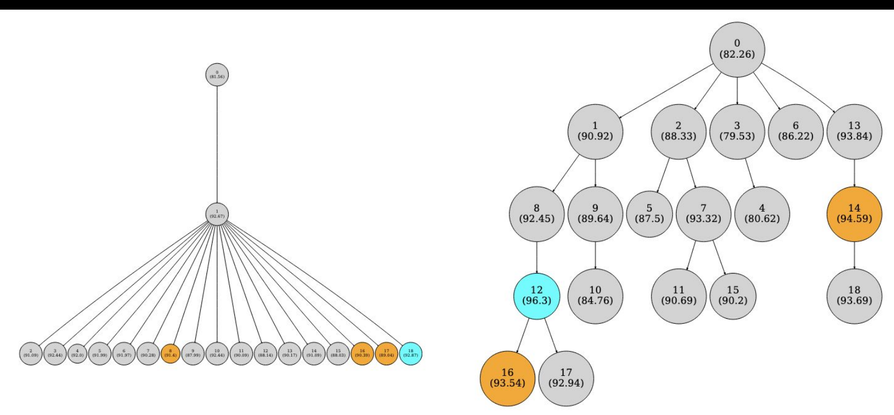

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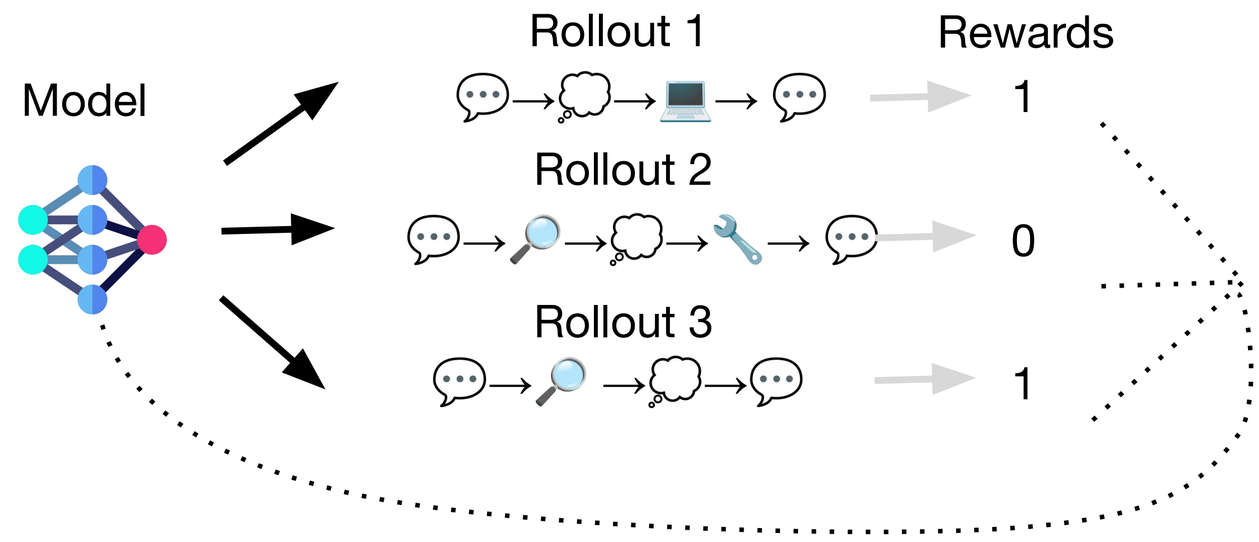

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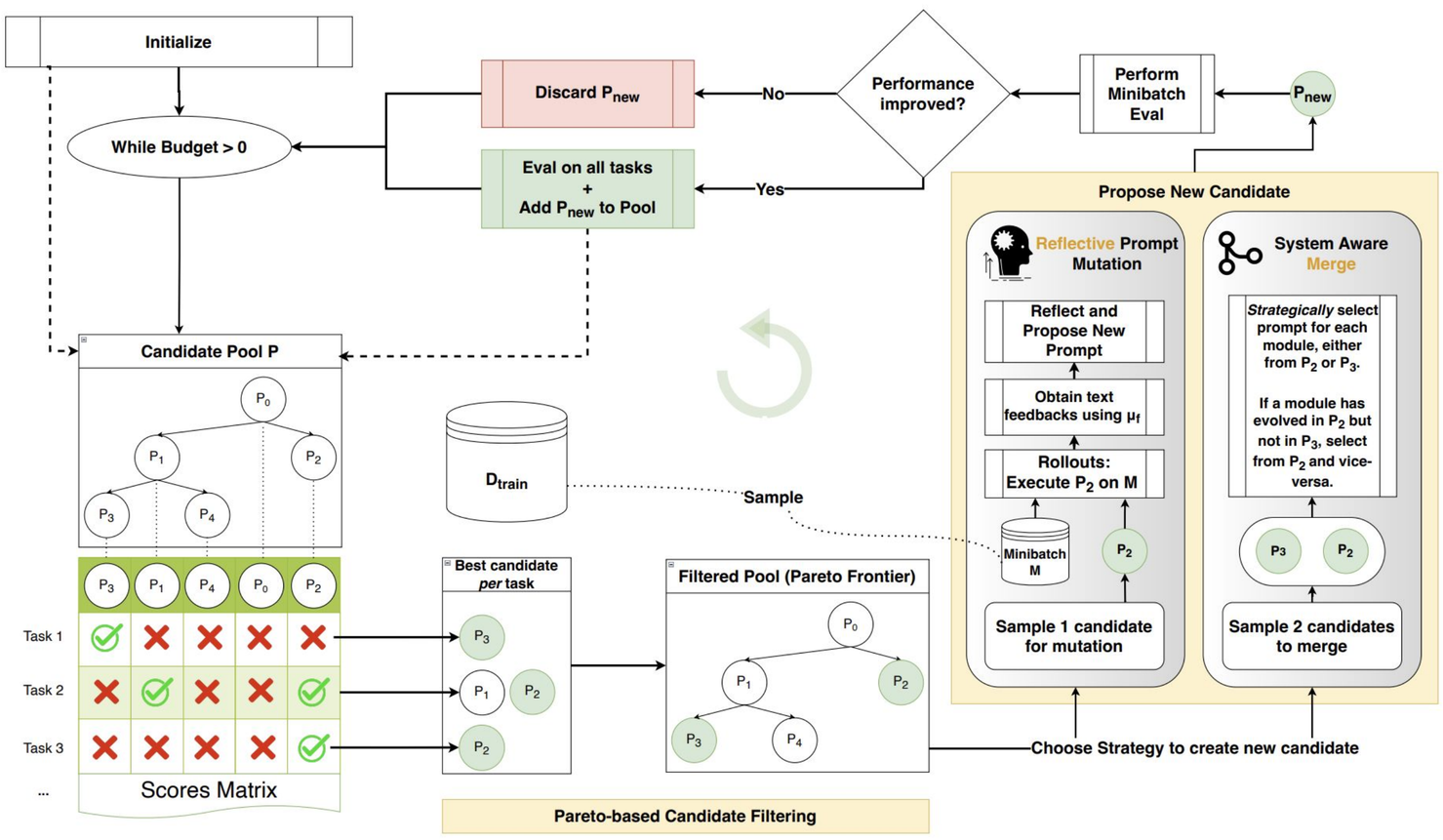

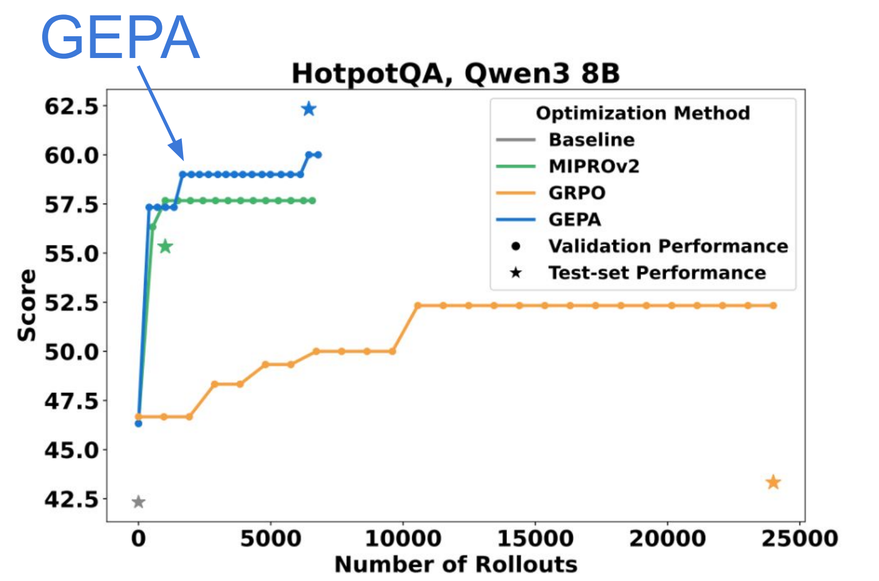

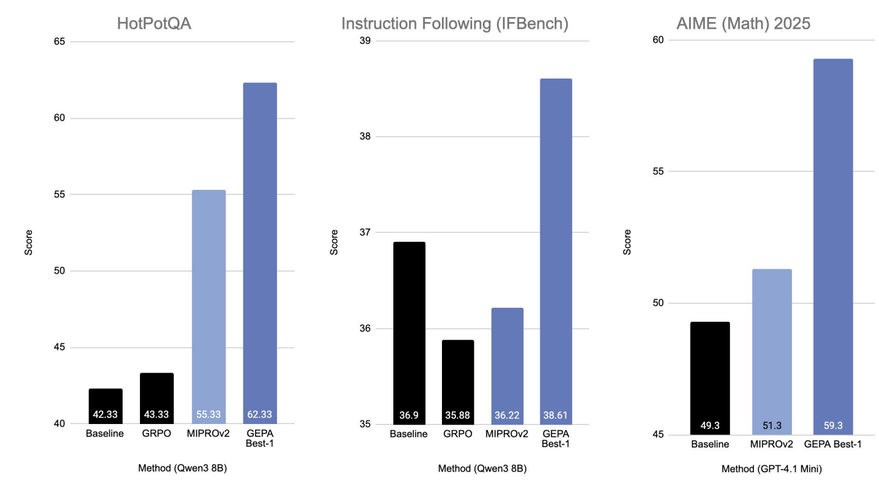

Seed Prompt for Second-Hop of Multi-Hop QA System

Given the fields `question`, `summary_1`, produce the fields `query`.

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Practical Strategy:

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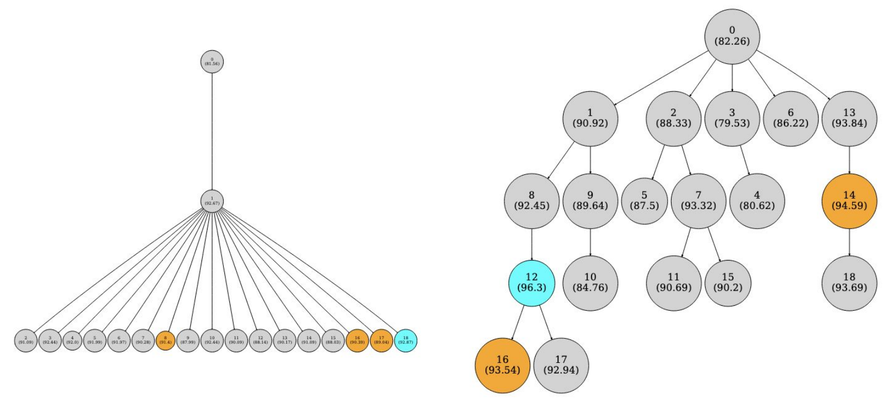
Qwen3 8B	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23	—
GRPO	43.33	35.88	38.67	86.66	**38.00**	51.26	48.91	+3.68
MIPROv2	55.33	36.22	47.33	81.55	20.00	46.60	47.84	+2.61
GEPA	62.33	**38.61**	**52.33**	**91.85**	32.00	**51.95**	**54.85**	**+9.62**
GEPA+Merge	**64.33**	28.23	51.67	86.26	32.00	**51.95**	52.40	+7.17
Total optimization budget (# rollouts)								
GEPA (+Merge)	6871	3593	7051	2426	1839	1839	3936	—
GRPO	24000	24000	24000	24000	24000	24000	24000	—

GPT-4.1 Mini	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	38.00	47.79	46.33	78.57	49.33	58.20	53.03	—
Trace (OptoPrime)	60.33	51.19	46.00	74.18	45.33	60.74	56.30	+3.27
MIPROv2-No-Demos	38.00	52.04	51.33	91.85	48.67	60.97	57.14	+4.11
MIPROv2	58.00	49.15	48.33	83.37	51.33	61.84	58.67	+5.64
TextGrad	62.33	48.64	47.67	85.68	46.67	63.84	59.14	+6.11
GEPA	**69.00**	52.72	51.67	94.47	**59.33**	**64.13**	65.22	+12.19
GEPA+Merge	65.67	**55.95**	**56.67**	**96.46**	**59.33**	**64.13**	**66.36**	**+13.33**
Optimized with Qwen3-8B, evaluated on GPT-4.1-Mini								
GEPA-Qwen-Opt	65.67	49.83	54.67	90.05	52.67	59.31	62.03	+9.00

Ablation: Importance of Candidate Selection



(a) SelectBestCandidate Strategy (b) Pareto-based candidate sampling.

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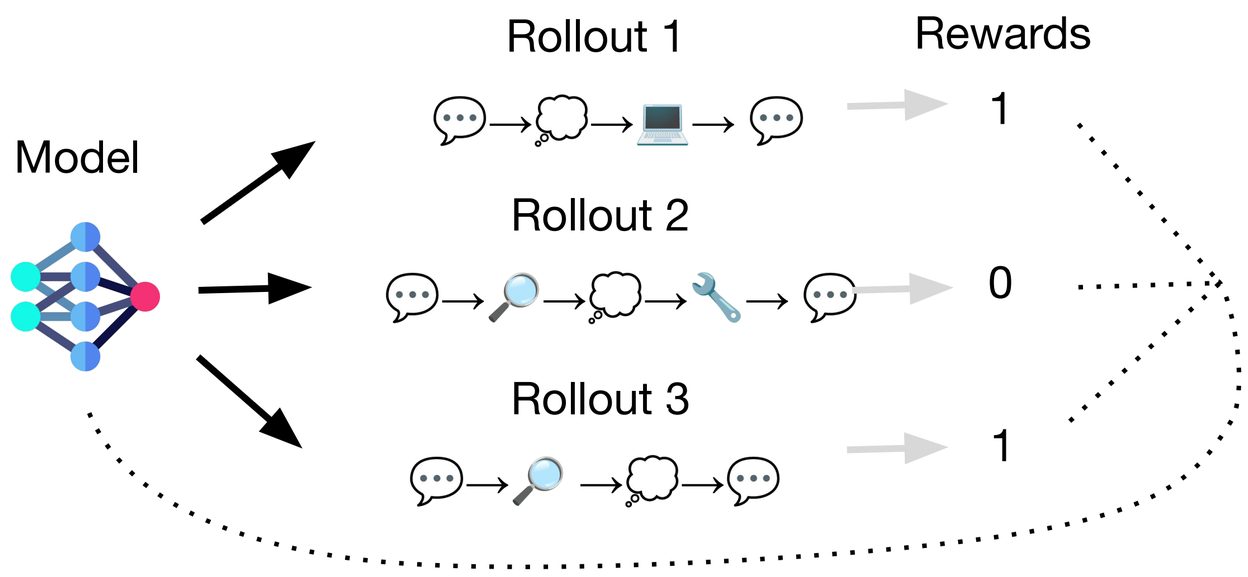
Motivation



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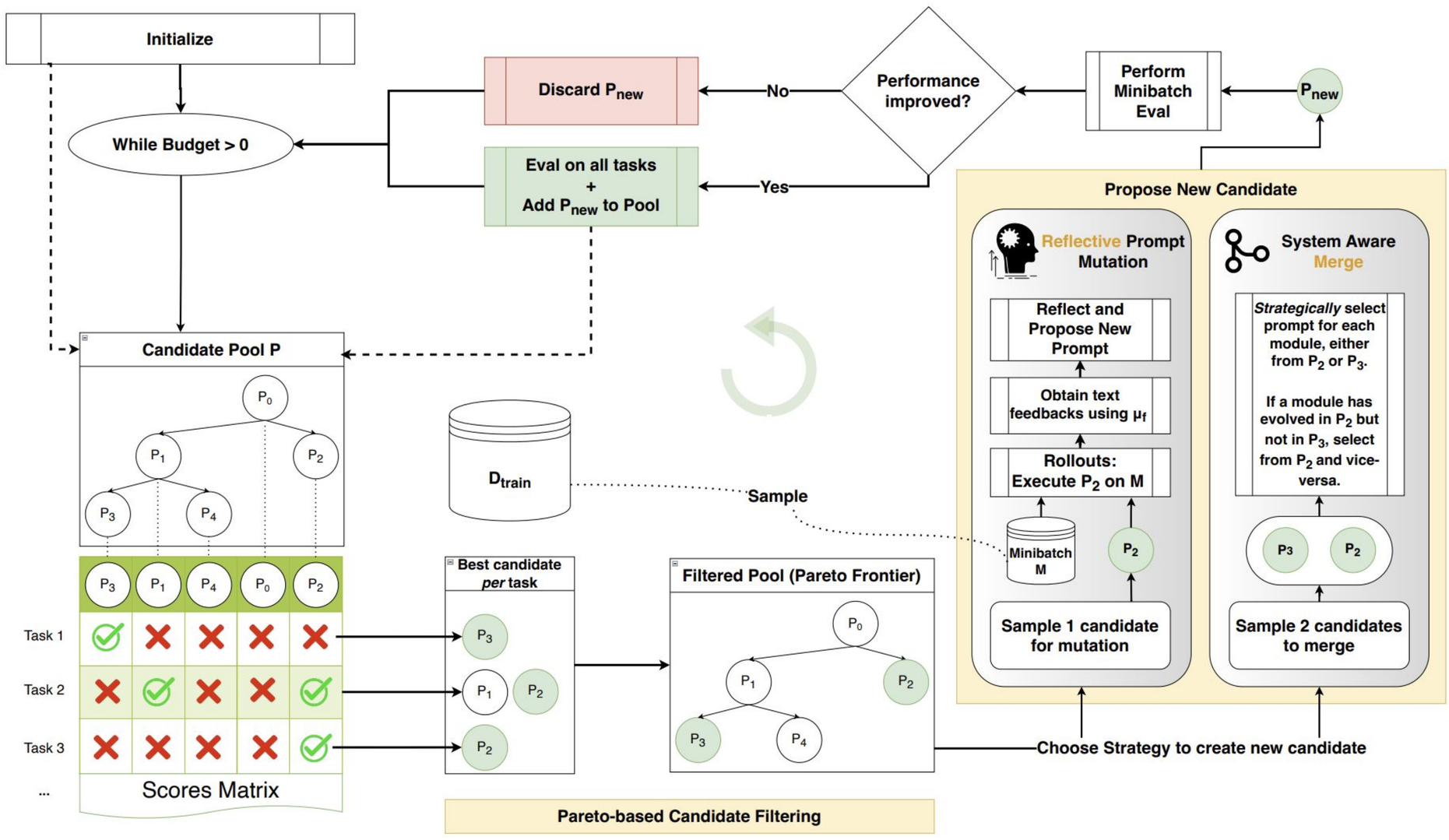

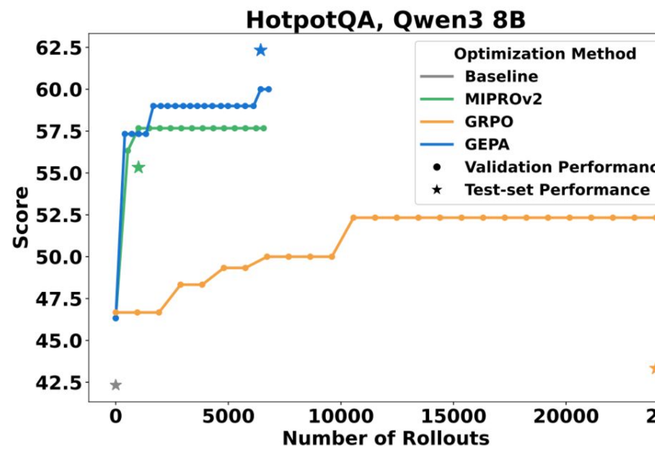

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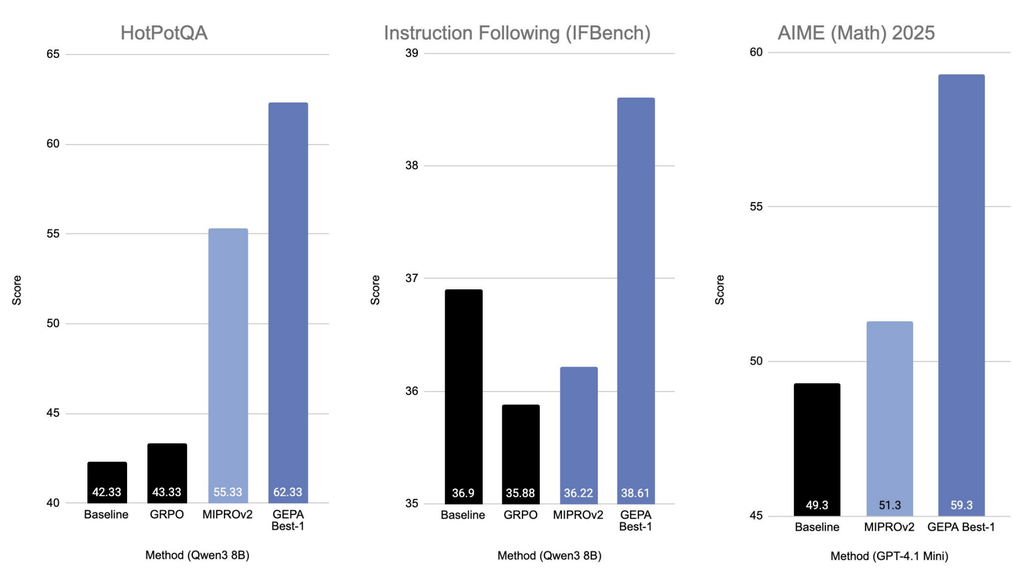

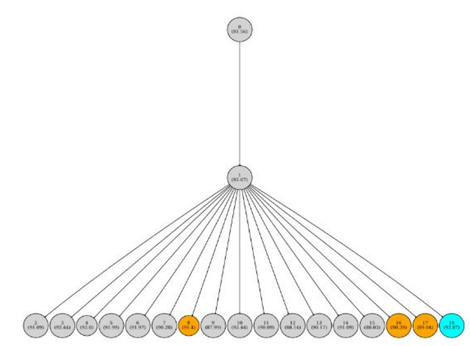

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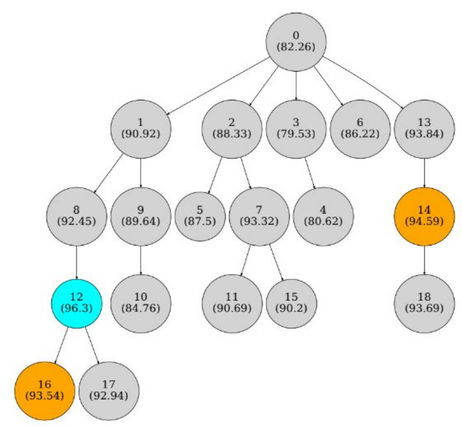

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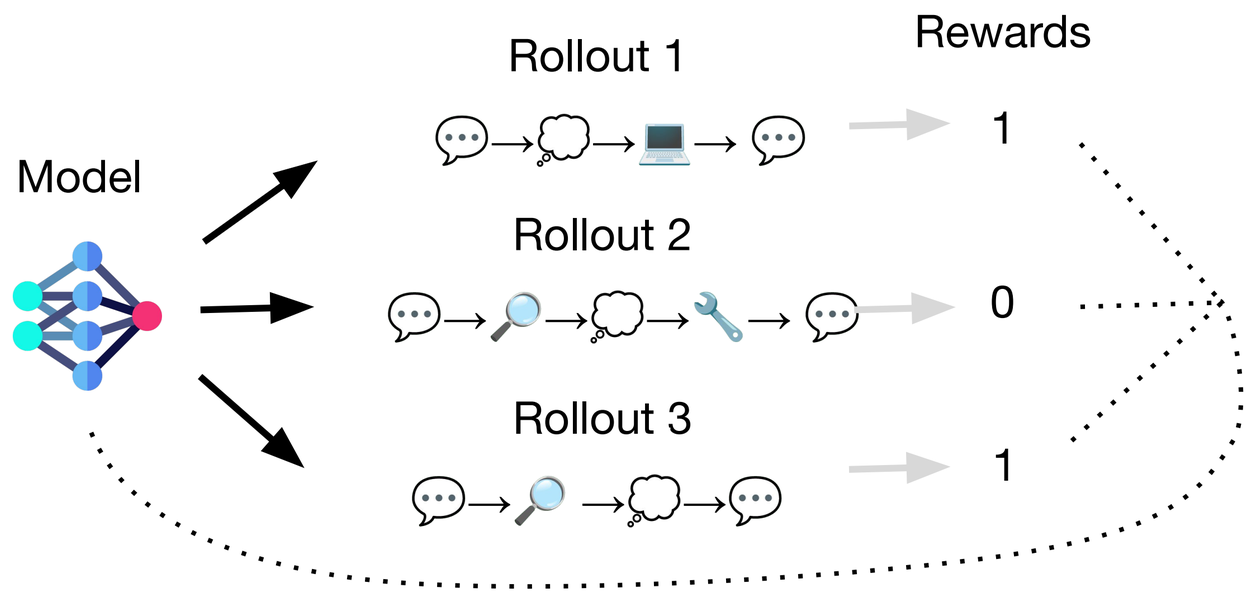

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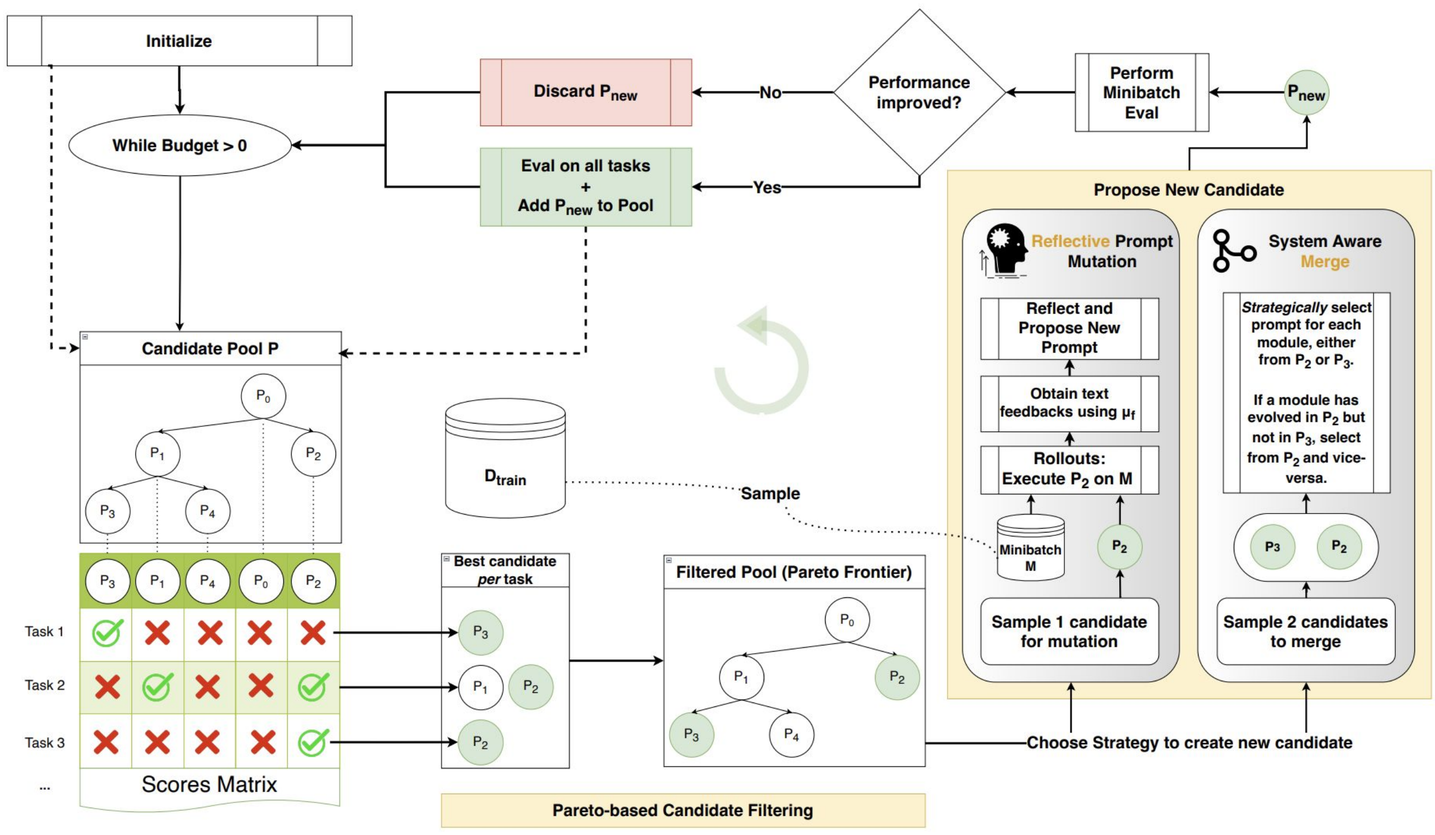
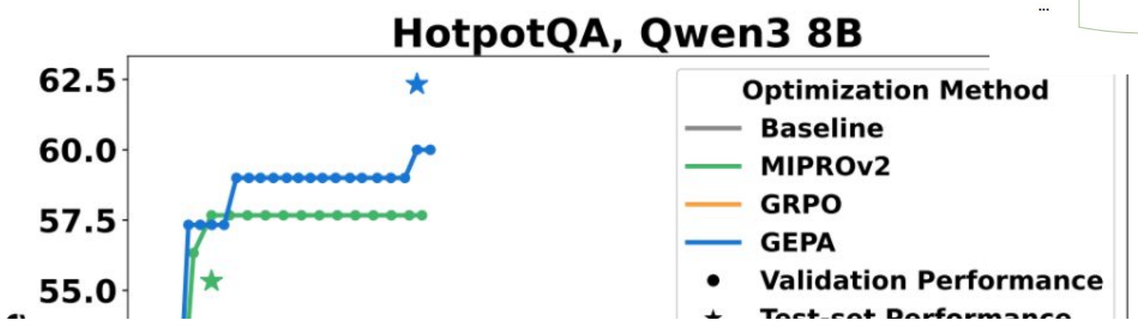
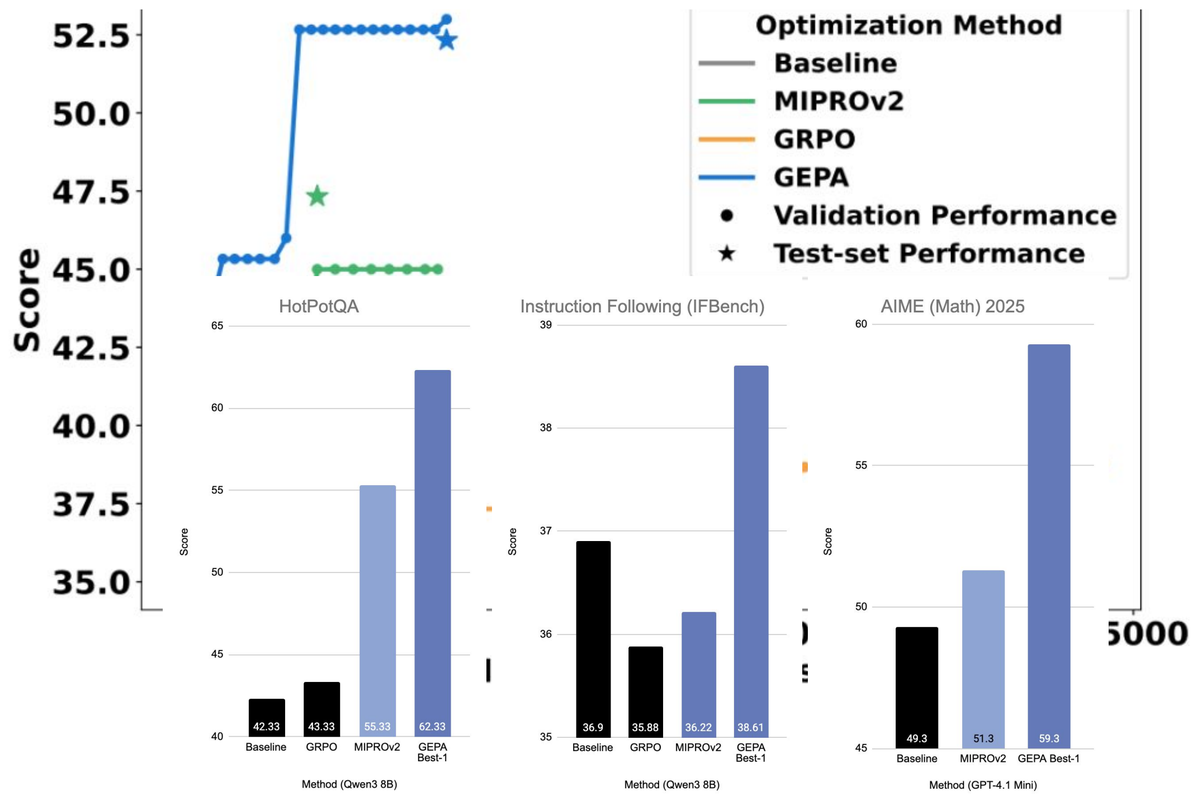
Insight: Text space

It is possible to use *very few* data and rollout samples, to reflect and extract new insights about the task.

Large and successful changes can be made to the model behaviour by updating the prompt.







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76% with GPT-5 Mini

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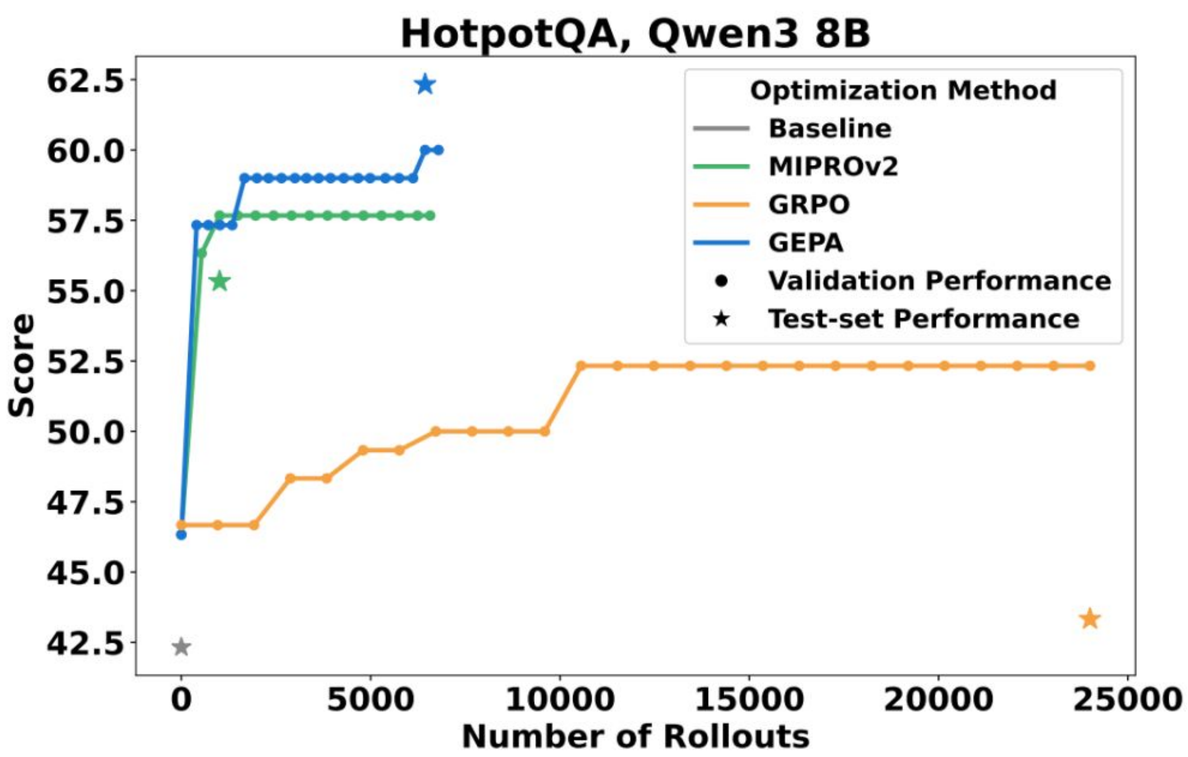

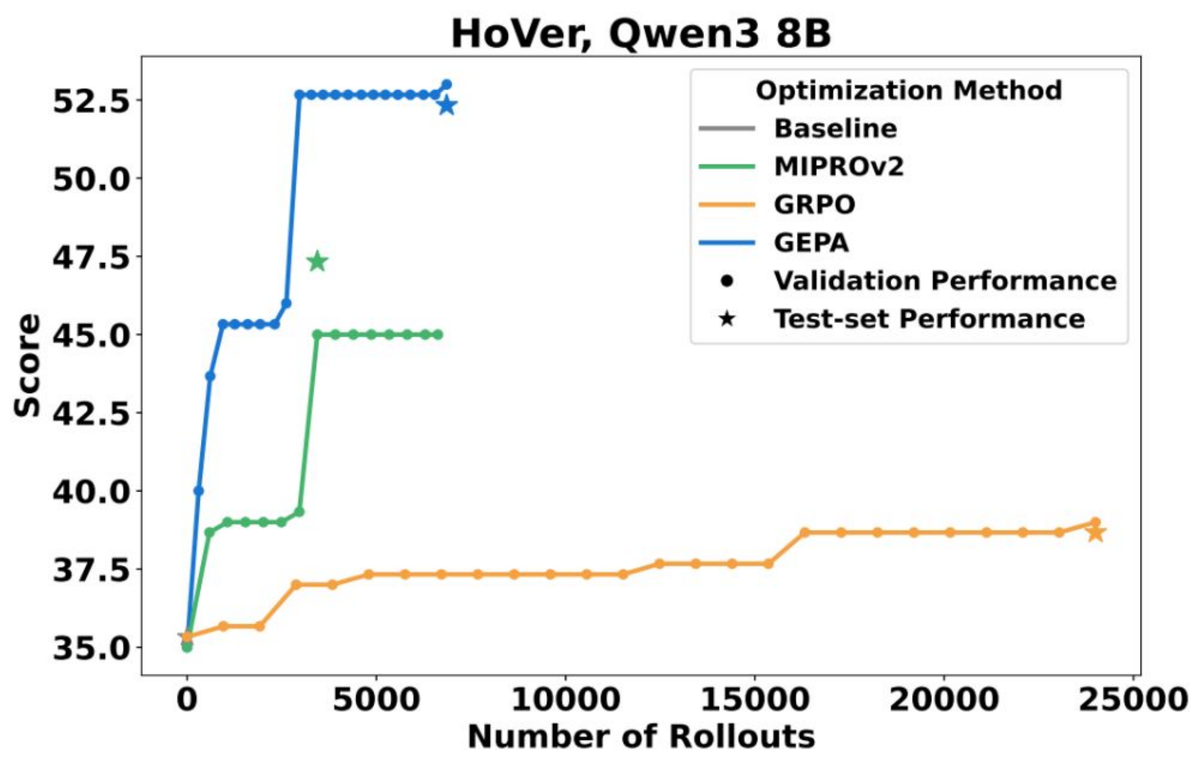

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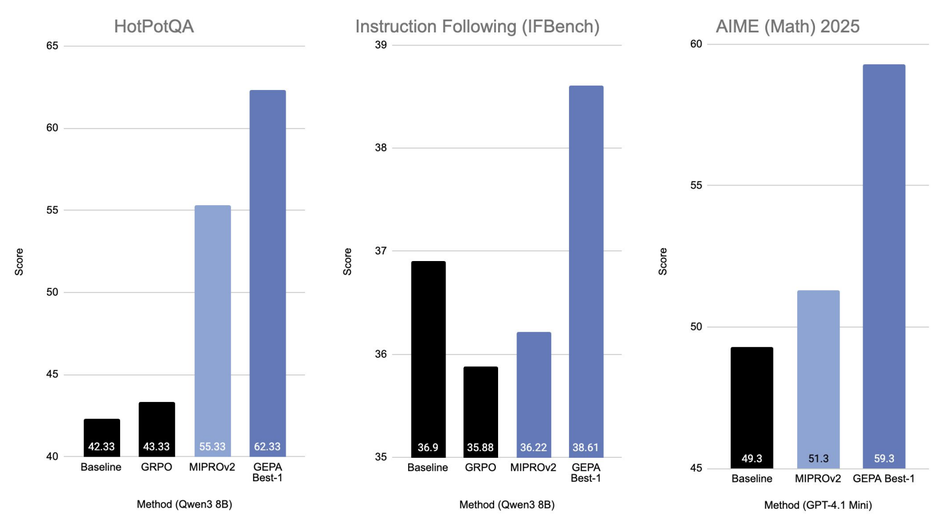

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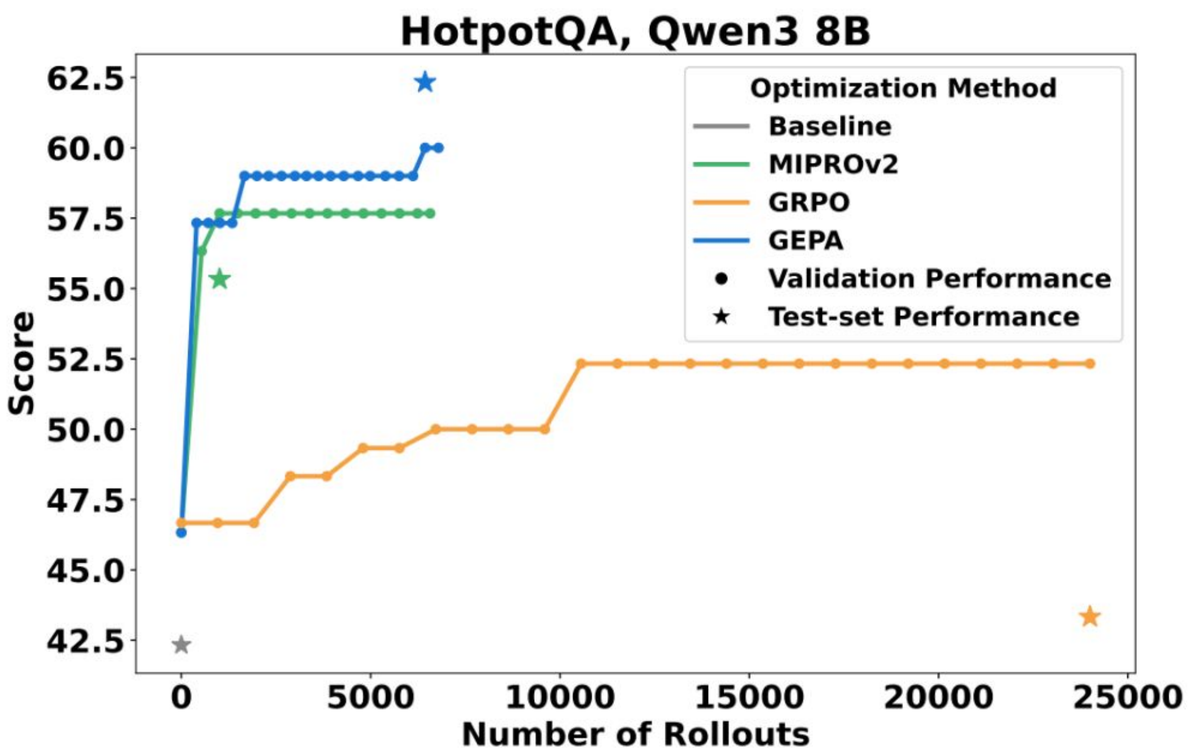

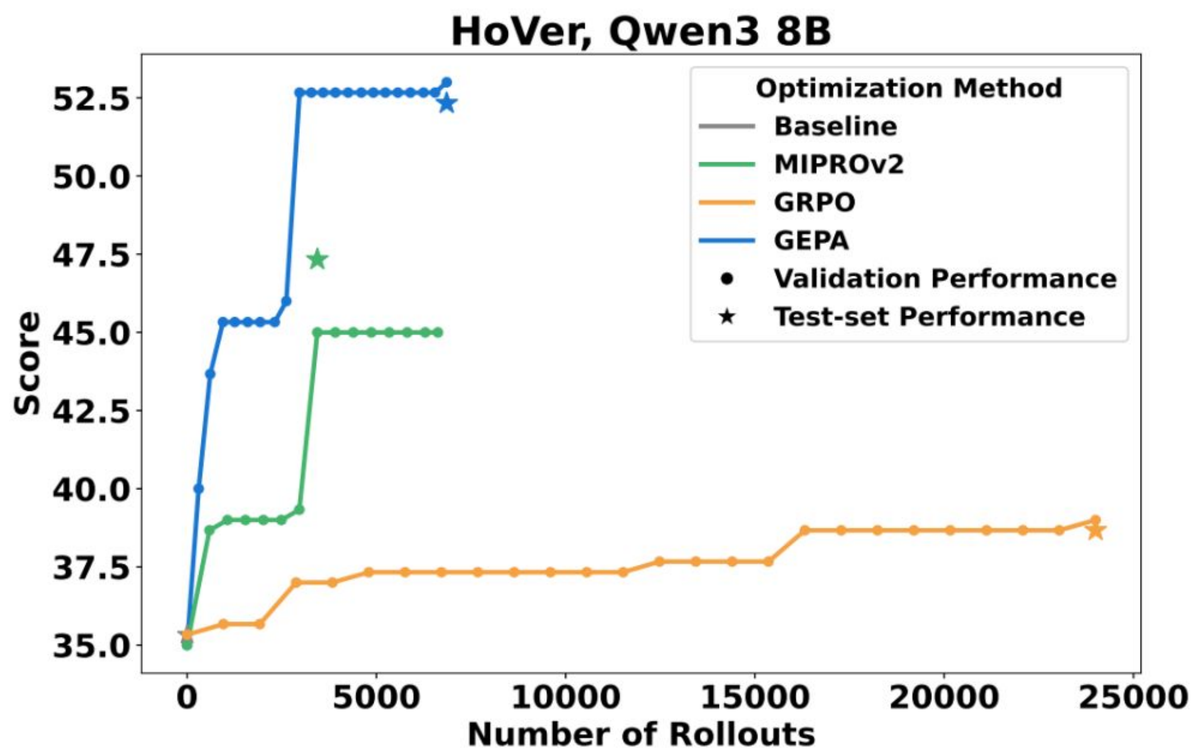

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- The query must retrieve relevant documents *NOT* found in first hop . . . for final answer extraction.

Key Observations and Lessons:

- First-hop documents often cover one entity or aspect.
- Remaining relevant documents often involve connected or higher-level concepts mentioned in `summary_1` but not explicitly asked in the original question. The `query` should target these *missing*, but logically linked, documents.
- Avoid merely paraphrasing the original question or restating known facts from `summary_1`.
- Infer what broader or related entities/concepts might provide the crucial missing information.
- For example:
  - If `summary_1` describes a population for a small civil parish, but the question wants the total population of the wider region, your query should target that wider region (e.g., "Madeira archipelago population in 2011").
  - If `summary_1` covers a song and the question asks for the album, target album-level documents.

How to Build the Query:

- Identify entities or topics mentioned in `summary_1` that are related but different from first-hop documents.
- Reframe the query to explicitly mention these broader or related entities *connected to the original question*.
- Include relevant key context from the question to maintain specificity, but shift focus to the missing piece.
- The goal is to retrieve documents that link or complement what was retrieved initially.

Practical Strategy:

- Read the `summary_1` carefully to spot references to bigger contexts or other entities not covered in the first hop.
- Ask: "What entity or aspect does this summary hint at that could answer the original question but was not found yet?"
- Formulate a precise, focused factual query targeting that entity or concept to retrieve the missing documents.

Output:

- Produce `query` as a clear, concise question or keyword phrase designed for efficient retrieval of second-hop documents.
- Ensure the query relates logically to the original question while targeting the broader or complementary knowledge identified in `summary_1`. . . . Do not include the original question or simply rephrase it. Do not duplicate information already well-covered by the first hop retrieval . . .